

FoodGenome AI

Calibrated Food Recognition with Conformal Guarantees
and Grounded Nutritional Retrieval

*What five negative results taught us about
when added complexity does not pay*

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Source, live demonstration and all evaluation artifacts:
<https://github.com/A-K-M-Asifuzzaman/Food> • <https://food-red-omega.vercel.app>

Abstract

We present FoodGenome AI, an end-to-end system that classifies a photographed dish into one of the 101 Food-101 categories, reports a calibrated confidence, returns a conformal prediction set carrying a measured coverage guarantee, and answers natural-language questions about the dish from a USDA-grounded knowledge base without stating any figure it cannot cite.

An ensemble of two frozen vision backbones reaches 97.16% top-1 on the held-out test split. Temperature scaling reduces expected calibration error 8.2-fold, and split-conformal prediction attains 99.56% empirical coverage at a mean set size of 1.54. A retrieval pipeline combining lexical and dense search answers 98.6% of a 76-case gold set correctly, refuses 100% of out-of-scope questions, and grounds every numeric claim against a retrieved source.

The contribution we consider most useful is negative. Five hypotheses that a practitioner would reasonably expect to hold did not: a trained fusion head lost to a parameter-free average; a self-supervised third backbone predicted to decorrelate proved the most correlated; attention-pooling maps localised the subject *worse than chance*; a conformal implementation whose calibration and deployment rules differed produced a better-looking number from a void guarantee; and end-to-end fine-tuning improved the ensemble by only 0.11 points, short of significance at $p = 0.052$. Each was established with an exact McNemar test or a measured baseline rather than by inspection.

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1 Introduction

Food recognition is an attractive application of computer vision because the output is immediately useful: a diner who photographs a plate wants to know what is on it and what it contains. It is also a domain where being confidently wrong carries real cost. A nutrition figure presented without qualification will be acted on, and a classifier restricted to 101 categories will return one of them for a photograph of anything at all.

This project therefore treats *knowing when not to answer* as a first-class requirement alongside accuracy. Three mechanisms implement that: a calibrated confidence whose numeric value corresponds to observed correctness, a conformal prediction set with a coverage guarantee that holds without distributional assumptions about the model, and an abstention rule that declines images the model cannot resolve. A fourth applies to the language side: a grounding gate that withholds any generated answer containing a quantity absent from its retrieved sources.

1.1 Contributions

- An ensemble reaching 97.16% Food-101 top-1 on a test split untouched until final evaluation, with model selection performed on a fixed validation slice carved from the training set.
- A calibration and conformal-prediction pipeline delivering a measured 99.56% coverage guarantee at a mean set size of 1.54, and an abstention rule that raises accuracy on accepted images to 97.94%.
- A retrieval-augmented question-answering system with a numeric grounding gate, evaluated on a gold set whose ground truth is derived rather than judged.
- Five negative results, each established by significance test or measured baseline, which together argue that added complexity in this setting frequently does not pay.

2 Method

2.1 Data and the validation protocol

Food-101 [1] contains 101,000 images across 101 classes, distributed as 75,750 training and 25,250 test images. It ships no validation split. Reporting model-selection numbers on the test set is the most common way work of this kind overstates itself, so a fixed 4% class-stratified slice (3,030 images, seed 1337) is carved from the training set and shared by every stage. The test split is untouched until final evaluation.

Because the fine-tuning stage ran on separate hardware, the validation mask is hashed and the hash asserted before training is permitted to begin. A divergent slice would make the two sets of results incomparable and would leak validation data into any ensemble of them. Both label arrays were later confirmed byte-identical across machines.

2.2 Frozen feature bank

Three backbones — SigLIP-SO400M [2], EVA-02-L [3] and DINOv2-L [4] — were run once over the full corpus and their pooled embeddings cached, a one-time cost of roughly twenty hours. Every subsequent experiment (head architecture, calibration, conformal sets, abstention, ensembling) then completes in seconds rather than hours. This is what made the ablation in Section 3.1 affordable: it was rerun dozens of times.

Lightweight MLP heads (1.7–1.9M parameters) are trained on the cached features with mixup and label smoothing. Features are L_2 -normalised before the head, which must be replicated at inference.

2.3 Reliability

Let $p_\theta(y | x)$ denote the ensemble’s output. A single scalar temperature T is fitted on the validation slice by minimising negative log-likelihood, giving calibrated probabilities $\text{softmax}(\log p_\theta/T)$. Temperature scaling is monotonic and so cannot alter accuracy; it alters only what the reported number means.

For conformal prediction we use split-conformal with the LAC score $s(x, y) = 1 - \hat{p}(y | x)$ [6]. Given miscoverage α , the threshold $\hat{q}$ is the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of calibration scores, and the prediction set is $\{y : \hat{p}(y | x) \geq 1 - \hat{q}\}$. The finite-sample correction is not cosmetic: at $n = 3030$ omitting it under-covers.

2.4 Retrieval and grounding

The knowledge base holds all 101 classes with 32 USDA nutrients each. USDA SR Legacy contains a direct record for 41; the remaining 60 are composed from weighted ingredient records, and every surface states which of the two a figure is.

Retrieval fuses BM25 and a bi-encoder by Reciprocal Rank Fusion [8] rather than by weighted score sum: BM25 scores are unbounded while cosine similarity is bounded, so any weighted combination requires a normalisation constant that is itself tuned. A cross-encoder then reranks. Crucially, the query is rewritten to name the dish the vision model has already identified — real questions contain pronouns (“how much sodium is in *this*”), which no retriever can resolve against 101 near-identical documents.

Generation is constrained to the retrieved context, and every quantity in the output is verified against that context before display, matched per unit so that milligrams cannot be supported by the same digits in grams. An answer that fails is withheld and a deterministic summary of the retrieved record served instead.

3 Results

3.1 Single models and the ablation

Table 1: Individual heads on the held-out test split. Frozen-feature performance tracks pretraining data rather than architecture: SigLIP’s caption supervision already encodes food semantics.

Head	Params	Val top-1	Test top-1	Test top-5
SigLIP-SO400M	1.93M	95.64	96.83	99.53
EVA-02-L	1.73M	93.14	95.53	99.30
DINOv2-L	1.73M	92.64	94.87	99.20
Gated fusion head	3.97M	–	96.97	99.48
EVA-02-L fine-tuned	304M	92.94	95.90	99.52

Negative result 1: a trained fusion head lost to arithmetic. A gated fusion head with 3.97M parameters, trained for 21.9 minutes to weight each backbone per input, reached 96.97%. A parameter-free 50/50 average of the same two probes reached 97.16% in milliseconds. An

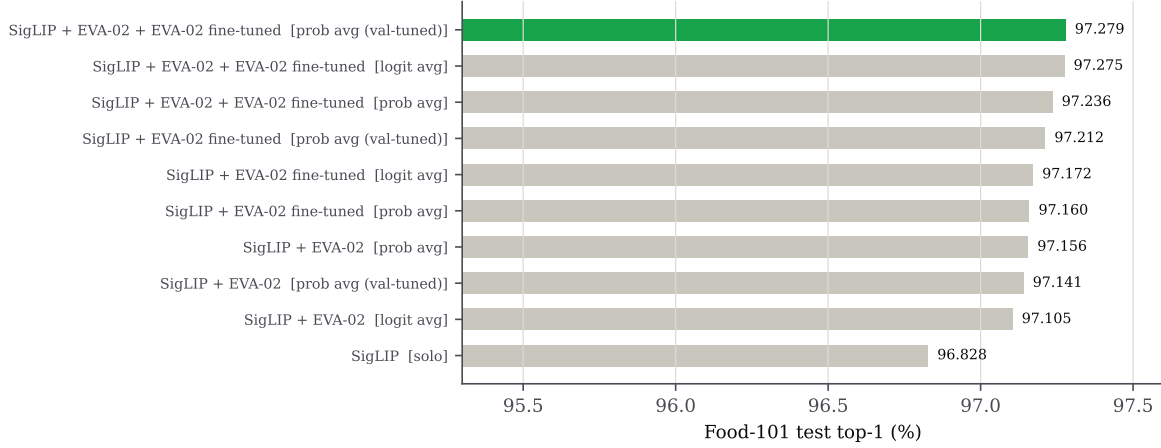
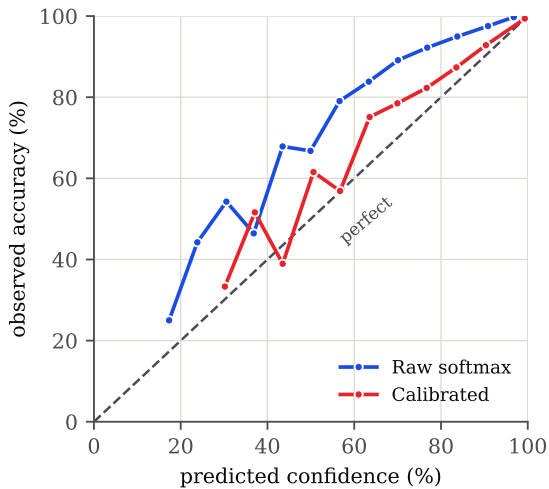


Figure 1: Every combination evaluated. The best configuration is a weighted probability average with no learned parameters.

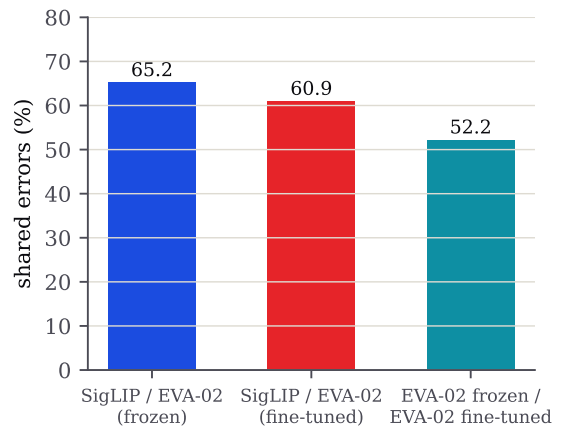
exact McNemar test on paired predictions puts the average ahead at $p = 0.005$, while the fusion head does *not* significantly beat its own best single input ($p = 0.071$). With two heavily correlated members, the additional capacity purchases variance rather than signal.

Negative result 2: the third backbone earned no place. DINOv2-L is the only self-supervised member and was therefore predicted to make the most independent errors. It makes the least: 66.3% of SigLIP’s errors are shared with it, against 65.2% with EVA-02. Adding it moved the three-way average to 97.09%, marginally below the pair, at $p = 0.312$ — not a significant loss, but no gain either. A validation weight search independently drove its weight to 0.05.

3.2 Calibration and conformal prediction



(a) Reliability diagram. The ensemble sat *above* the diagonal — systematically under-confident, the less common direction.



(b) Shared-error rate. Fine-tuning at 448px is what produced diversity; at 224px it produced none.

Figure 2: Reliability and diversity.

Temperature scaling fitted $T = 0.7621$, reducing expected calibration error from 0.0514 to 0.0062 — a factor of 8.2. Mean confidence moved from 92.0% to 96.6% against 97.16%

accuracy; the two now agree.

Table 2: Conformal coverage against mean set size. Below the model’s own top-1 accuracy the guarantee is satisfied by the single best prediction and the “set” degenerates to one class.

Target	Method	Coverage	Mean size	Singletons
90%	LAC	93.59	0.95	94.6%
90%	APS	90.18	1.54	80.4%
95%	APS	95.56	2.52	79.7%
99%	LAC + top-1	99.56	1.54	75.9%

Negative result 3: a mismatched conformal rule looked better. An initial implementation forced the top-1 class into the prediction set at test time while scoring *without* it during calibration. This silently rescued the 13% of images whose top score exceeded the threshold, producing 99.1% empirical coverage against a 90% target. The guarantee was void, and the reported number was *better* for it — which is what makes this class of error dangerous. Conformal validity requires the identical rule in calibration and deployment; after correction, APS attains 90.18% against a 90% target.

Abstention. Neither available signal suffices alone. A maximum-probability threshold rejects noise but admitted a flat colour field at 0.43, classified “breakfast burrito”. Conformal set size caught that at eight candidates where correctly classified dishes return one. Their disjunction abstains on 2.28% of real food and raises accuracy on the remainder from 97.16% to 97.94%. This is an abstention rule, not a trained out-of-distribution detector: it reports that the model is lost, which correlates with but does not equal “this is not food”.

3.3 Explainability

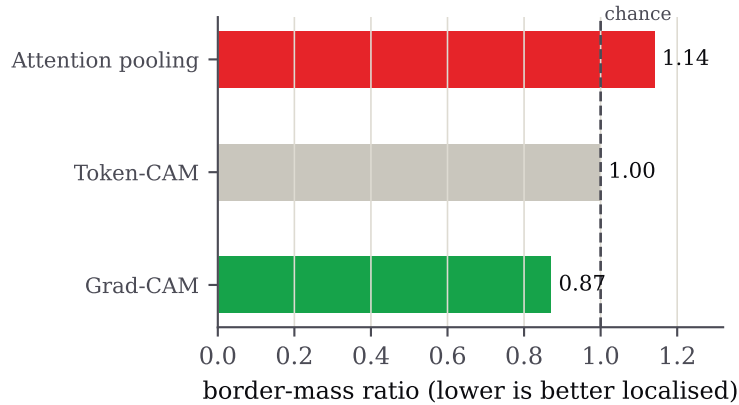


Figure 3: Attribution localisation, measured as the share of attribution mass falling in the outer ring of the frame (60% of the area) over 24 test images. Food-101 subjects are typically centred, so lower is better.

Negative result 4: attention is not saliency. Reading the pooling head’s own attention weights ought to be the most faithful account of where the model looked — those weights are literally the mixture forming the classified vector. Measured against a border-mass baseline, it is *worse than chance* at locating the subject ($1.14\times$), while classic Grad-CAM [9] reaches $0.87\times$

and is also far more class-discriminative (map correlation $r = +0.07$ between two target classes, against $r = +0.38$ for token-level attribution). This reproduces the documented tendency of vision transformers to place high attention on low-information background patches [10].

Localisation is weak in absolute terms for all three methods, and the cause is architectural: the classifier sees only a globally pooled vector, so spatial attribution must be recovered through a step that discarded position.

3.4 Fine-tuning

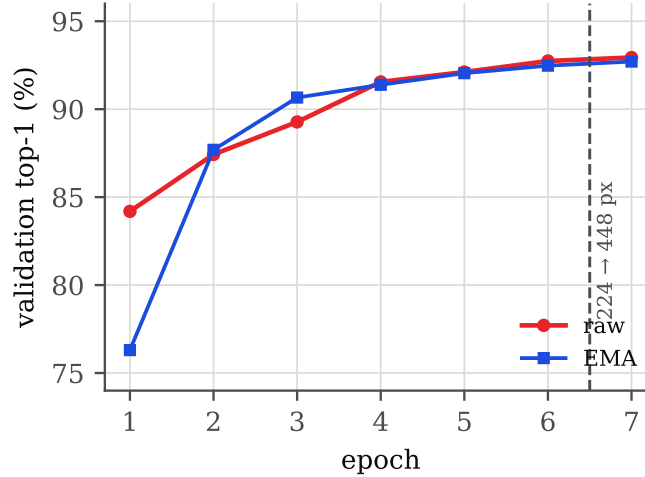


Figure 4: Fine-tuning EVA-02-L: six epochs at 224 px followed by one at 448 px. The resolution change contributes 0.2 points of validation accuracy.

Negative result 5: fine-tuning decorrelated but did not significantly win. End-to-end fine-tuning reached 95.90% alone — below the frozen ensemble. Its value was expected to lie in diversity, and at 224 px there was none: 65.5% of SigLIP’s errors shared, indistinguishable from frozen EVA-02’s 65.2%. The single 448 px epoch changed that, dropping shared errors to 60.9% (Figure 2b) while adding only 0.2 points of validation accuracy. Its contribution was diversity, which no accuracy column reveals.

The resulting three-way ensemble reaches 97.27%, an improvement of 0.11 points that does not clear significance ($p = 0.052$; 111 images gained, 83 lost). Given that serving it requires a 304M-parameter model against two small heads over cached features, the frozen pair remains the deployed configuration.

3.5 Retrieval and grounding

Overall: 98.6% of in-scope questions answered correctly, 100% refusal accuracy with zero false refusals, 100% of answers grounded, at a total evaluation cost of \$0.024. Ground truth is derived from the knowledge base rather than judged by a model, since a language-model judge would share error modes with the system under test.

Strict recall@1 reads 64.6% on nutrient lookups. This is a labelling artefact: every miss returned a *portion* document where a *macros* document was labelled, and all 17 contained the correct figure. An answerability metric, measuring whether the top hit contains the requested value, reads 100%.

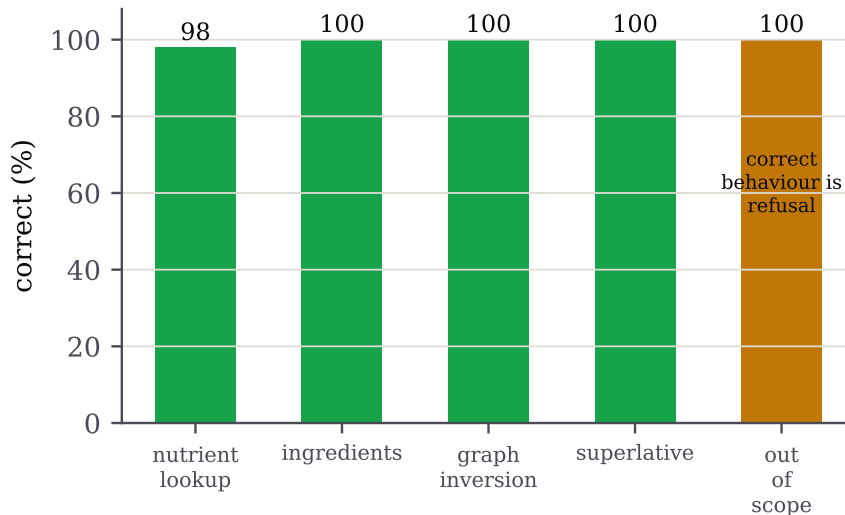


Figure 5: Answer correctness by question category on the 76-case gold set. For out-of-scope questions the correct behaviour is refusal, scored separately so that a fluent answer is counted as the failure it is.

The grounding gate caught a genuine hallucination on its first live query. Asked for per-serving sodium, the model produced 983.4 mg — arithmetically correct (447×2.2) and derivable from a stated multiplier, but present in no document. The correction was made to the corpus rather than the tolerance: micronutrient documents now state both bases explicitly, removing the need to derive.

4 Discussion

Four of the five negative results share a shape: an added component that should have helped did not, because the quantity it was meant to improve was already saturated. The fusion head had nothing to learn because two correlated members need no per-input weighting. DINOV2 added no diversity because self-supervision did not produce independent errors on this distribution. Fine-tuning added diversity but too little to move a 25,250-image comparison.

The oracle accuracy — the ceiling if a perfect combiner always selected a correct member — is 98.43%. Only 1.57% of test images defeat every model, and inspection suggests a substantial fraction are Food-101 label noise. Ensemble-based approaches on this dataset are close to exhausted.

4.1 Limitations

- Attribution localises weakly ($0.87\times$ chance) for architectural reasons described above; the maps indicate a region, not an object.
- Sixty of the 101 nutrition profiles are composed from weighted ingredients rather than measured directly, and carry correspondingly more uncertainty.
- Abstention is a baseline rule, not a trained out-of-distribution detector, which would require non-food negatives.
- The knowledge base is closed at 101 categories; anything outside is refused rather than estimated.

5 Conclusion

FoodGenome AI classifies at 97.16% on a sealed Food-101 test split, reports confidences that match observed accuracy to within 0.6 points of calibration error, supplies prediction sets with a measured 99.56% coverage guarantee, and answers grounded nutritional questions while refusing those it cannot support.

The five negative results are the part we would emphasise. Each concerns a component that a practitioner would reasonably expect to help, each was tested rather than assumed, and each failed. Reporting them costs a headline number and buys something more useful: a defensible account of where the remaining headroom is not.

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